Nonexistence of the proposed persistence density

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Abstract

For every fixed integer sample depth $N \geq 3$, the persistent integer bases defined by Hyman's continued-fraction criterion have no natural density. Thus the proposed density 1/12 does not exist. The proof uses two open intervals of fractional logarithms, one entirely persistent and one entirely nonpersistent, and their repetition in every decimal decade.

Task 1: the precise density question

Fix $N \geq 3$ and write log for the natural logarithm. For an integer $b \geq 2$ with irrational $\alpha_b = \log_{10} b$, write $\alpha_b = [a_0; a_1, a_2, \ldots]$, and let q_k be its convergent denominators, with $q_{-1} = 0$, $q_0 = 1$. Following Corollary 3.16 of [1], define

$$\mathcal{P}_N = \{b \geq 2 : \text{ some } k \geq 0 \text{ has } q_k \leq N \text{ and } a_{k+1} q_k > N \log N\}.$$

The rational-logarithm bases may be included or excluded arbitrarily; they have density zero, as shown below. Conjecture 3.17 proposes

$$\lim_{B \to \infty} \frac{\#(\mathcal{P}_N \cap [2, B])}{B - 1} = \frac{1}{12}.$$

The sample depth is retained explicitly here: the source defines persistence at depth N and surveys $N = 10{,}000$, with an extended survey at $N = 200{,}000$. Its supplementary Section 9.8 also keeps N fixed in the continued-fraction cutoff while increasing the base bound.

Theorem 1. *For every fixed integer $N \geq 3$, the natural density of $\mathcal{P}_N$ does not exist. In particular, Conjecture 3.17 fails for every such choice of the fixed sample depth, including both survey depths.*

Two intervals with opposite classifications

Adding an integer to a continued fraction changes only a_0. The recurrence $q_k = a_k q_{k-1} + q_{k-2}$ therefore shows that the persistence criterion depends only on $\theta = \{\log_{10} b\} \in (0, 1)$.

First choose $A = N^2$. Since $\log N < N$, we have $A > N \log N$. For every irrational

$$\theta \in J_+ := \left(\frac{A}{A+1}, 1 \right),$$

one has $a_1 = 1$, $q_1 = 1$, and

$$a_2 = \left\lfloor \frac{\theta}{1 - \theta} \right\rfloor \geq A > N \log N.$$

Thus every such θ meets the persistence criterion already at $k = 1$. This argument does not require admitting the index $k = 0$.

Next consider a continued-fraction prefix consisting entirely of ones. Let

$$p_{-1} = 1, \quad p_0 = 0, \quad q_{-1} = 0, \quad q_0 = 1, \qquad p_j = p_{j-1} + p_{j-2}, \quad q_j = q_{j-1} + q_{j-2}.$$

Choose L with $q_L > N$. The values

$$[0; \underbrace{1, \ldots, 1}_{L \text{ entries}}, t] = \frac{p_L t + p_{L-1}}{q_L t + q_{L-1}}, \qquad 1 < t < \infty,$$

fill the open interval J_- between

$$u = \frac{p_L}{q_L}, \qquad v = \frac{p_L + p_{L-1}}{q_L + q_{L-1}}.$$

The determinant $p_L q_{L-1} - p_{L-1} q_L = (-1)^{L-1}$, proved immediately by induction, gives

$$|u - v| = \frac{1}{q_L(q_L + q_{L-1})} > 0.$$

Every irrational number in J_- has $a_1 = \cdots = a_L = 1$. Since its Lth denominator exceeds N, every index with $q_k \le N$ has $k < L$. Consequently

$$a_{k+1} q_k = q_k \le N < N \log N,$$

where $\log N > 1$ for $N \ge 3$. None of these numbers is persistent.

Decimal intervals contradict a natural density

If an integer $b \ge 2$ has $\log_{10} b = p/q$ in lowest terms, then $b^q = 10^p$. Taking the exponent of the prime 2 gives $q\, v_2(b) = p$, so $q = 1$. Such bases are exactly the integral powers of ten. There are only $O(\log B)$ of them up to B.

Put

$$c_+ = 10^{A/(A+1)}, \quad d_+ = 10, \qquad c_- = 10^{\min(u,v)}, \quad d_- = 10^{\max(u,v)}.$$

For every nonnegative integer m, all integer bases in $(c_+ 10^m, d_+ 10^m)$ are persistent, while none in $(c_- 10^m, d_- 10^m)$ is persistent. These open intervals lie within a single decimal decade, so contain no powers of ten.

Suppose a natural density δ existed and set $C(B) = \#(\mathcal{P}_N \cap [2, B])$. Equivalently, $C(B) = \delta B + o(B)$. For fixed $1 \le c < d \le 10$, subtracting this asymptotic at the two endpoints gives

$$\#\big(\mathcal{P}_N \cap (c10^m, d10^m)\big) = \delta(d - c)10^m + o(10^m). \tag{1}$$

The possible integer endpoints change this equation by at most two. For $(c, d) = (c_+, d_+)$ the left side is $(d_+ - c_+)10^m + O(1)$, so (1) forces $\delta = 1$. For $(c, d) = (c_-, d_-)$ it is zero, forcing $\delta = 0$. This contradiction proves the theorem.

This is a refutation of the ordinary base-counting limit stated in the source. A logarithmically weighted density or a joint limit with $N = N(B)$ would be a different mathematical question. The proof uses the source's exact continued-fraction definition and does not require any claimed CMI estimates or survey classifications.

Reproducible checks

The accompanying exact verifier checks the continued-fraction recurrences, cylinder endpoint identities, and all relevant denominator cutoffs for every integer $3 \le N \le 10{,}000$, and also for $N = 200{,}000$ and $N = 10^9$. It verifies $1 < \log 3$ by a rational lower Taylor bound and uses $\log N \le N - 1 < N$ analytically. At $N = 3$, the inequalities $9^{10} > 10^9$, $9 < 10$, $4^5 > 10^3$, and $4^8 < 10^5$ place bases 9 and 4 in the two respective cylinders without floating-point logarithms. These checks supplement the universal proof.

Proposed follow-up tasks

Task 2. Determine the lower and upper natural densities of $\mathcal{P}_N$ at the survey depth $N = 10{,}000$.

Task 3. Determine whether $\mathcal{P}_N$ has a logarithmic density and express it in terms of continued-fraction cylinders.

Task 4. Determine the asymptotic behavior, as $N \to \infty$, of the Lebesgue measure of persistent rotation numbers in $(0, 1)$.

Task 5. Identify growth regimes $N = N(B)$ for which the persistence proportion up to base B has a limit, and compute it.

References

[1] J. M. Hyman (Mac Hyman), *Why Eight Percent of Benford Sequences Never Converge*, arXiv:2603.18243v1 (2026). Definition 3.11, Corollary 3.16 and Conjecture 3.17, pp. 16–18; Supplementary Information, Section 9.8, pp. 33–34, author repository.